PIC Code OOP Refactor — Final Directory Structure & Routine Mapping

This document presents the final directory layout for the modern, object-oriented PIC code and a one-to-one mapping of all your original routines to their new module/submodule homes.

Conventions:
- modules/ contain headers & public interfaces (types + procedure declarations).
- submodules/ contain the actual implementations (bodies) for those interfaces.
- Naming: module mod_X ↔ submodule (mod_X) submod_X

Final Directory Structure

```
./
├── CMakeLists.txt
└── Src
    ├── main.f90
    │
    ├── modules/                      # Interfaces (module headers)
    │   ├── core/
    │   │   ├── mod_kinds.f90
    │   │   ├── mod_constants.f90
    │   │   ├── mod_logger.f90
    │   │   ├── mod_timer.f90
    │   │   ├── mod_rng.f90
    │   │   └── mod_config.f90
    │   ├── domain/
    │   │   ├── mod_geometry.f90
    │   │   └── mod_boundary.f90
    │   ├── field/
    │   │   ├── mod_field.f90
    │   │   ├── mod_bfield.f90
    │   │   └── mod_poisson.f90
    │   ├── particles/
    │   │   ├── mod_species.f90
    │   │   ├── mod_injector.f90
    │   │   ├── mod_collisions.f90
    │   │   └── mod_reactions.f90
    │   ├── io/
    │   │   ├── mod_io.f90
    │   │   └── mod_restart.f90
    │   ├── diag/
    │   │   └── mod_diagnostics.f90
    │   └── driver/
    │       ├── mod_pic.f90
    │       └── mod_factory.f90
    │
    └── submodules/                   # Implementations (submodules)
        ├── core/
        │   ├── submod_kinds.f90
        │   ├── submod_constants.f90
        │   ├── submod_logger.f90
        │   ├── submod_timer.f90
        │   ├── submod_rng.f90
        │   └── submod_config.f90
        ├── domain/
```

Final Directory Structure (cont.)

```
│    ├── submod_geometry.f90
│    └── submod_boundary.f90
├── field/
│    ├── submod_field.f90
│    ├── submod_bfield_gaussian.f90
│    ├── submod_bfield_map.f90
│    ├── submod_poisson_mg.f90
│    ├── submod_poisson_sor.f90
│    └── submod_poisson_jacobi.f90
├── particles/
│    ├── submod_species.f90
│    ├── submod_injector.f90
│    ├── submod_collisions.f90
│    └── submod_reactions.f90
├── io/
│    ├── submod_io.f90
│    └── submod_restart.f90
├── diag/
│    └── submod_diagnostics.f90
└── driver/
     ├── submod_pic.f90
     └── submod_factory.f90
```

Mapping — Bfield.f90

```
read_Bfield_map              → modules/field/mod_bfield.f90  →
submodules/field/submod_bfield_map.f90
gaussian_Bfield              → modules/field/mod_bfield.f90  →
submodules/field/submod_bfield_gaussian.f90
PG_current                   → modules/field/mod_bfield.f90  →
submodules/field/submod_bfield_gaussian.f90
EE_magnets                   → modules/field/mod_bfield.f90  →
submodules/field/submod_bfield_gaussian.f90
Bfield_Hall_thruster         → modules/field/mod_bfield.f90  →
submodules/field/submod_bfield_gaussian.f90
Bx_magnet, By_magnet         → modules/field/mod_bfield.f90  →
submodules/field/submod_bfield_gaussian.f90
find_B_dir                   → modules/field/mod_bfield.f90  → (utility) implemented in
submodules/field/*
```

Mapping — calc_rho.f90

```
calc_rho        → modules/field/mod_field.f90   → submodules/field/submod_field.f90
dens_red        → modules/field/mod_field.f90   → submodules/field/submod_field.f90
```

Mapping — collisions.f90

```
collisions                     → modules/particles/mod_collisions.f90 →
submodules/particles/submod_collisions.f90
collision_OMP                   → modules/particles/mod_collisions.f90 →
submodules/particles/submod_collisions.f90
scatter                         → modules/particles/mod_collisions.f90 →
submodules/particles/submod_collisions.f90
```

Mapping — Efield.f90

calc_Efield → modules/field/mod_field.f90 → submodules/field/submod_field.f90

Mapping — generate_boundary.f90

generate_boundary → modules/domain/mod_boundary.f90 → submodules/domain/submod_boundary.f90

Mapping — indexx.f90

```
indexx                              → modules/particles/mod_reactions.f90 (or utils) →
submodules/particles/submod_reactions.f90
```

Mapping — load_part.f90

load_part → modules/particles/mod_injector.f90 → submodules/particles/submod_injector.f90
load_part_OMP → modules/particles/mod_injector.f90 → submodules/particles/submod_injector.f90
load_gauss → modules/particles/mod_injector.f90 → submodules/particles/submod_injector.f90

Mapping — mg.f90

```
mg                             → modules/field/mod_poisson.f90 →
submodules/field/submod_poisson_mg.f90
restriction, getres, prolongation → modules/field/mod_poisson.f90 →
submodules/field/submod_poisson_mg.f90
```

Mapping — part_expmover.f90

```
part_mover                    → modules/particles/mod_species.f90 →
submodules/particles/submod_species.f90
eheating                      → modules/particles/mod_species.f90 →
submodules/particles/submod_species.f90
charge_deposition             → modules/particles/mod_species.f90 →
submodules/particles/submod_species.f90
```

Mapping — part_flux_injection.f90

```
part_flux_injection          → modules/particles/mod_injector.f90 →
submodules/particles/submod_injector.f90
load_flux_OMP                 → modules/particles/mod_injector.f90 →
submodules/particles/submod_injector.f90
```

Mapping — part_injection.f90

part_injection → modules/particles/mod_injector.f90 → submodules/particles/submod_injector.f90
shifted_maxwellian_flux → modules/particles/mod_injector.f90 → submodules/particles/submod_injector.f90

Mapping — pdesolver.f90

```
pdesolver                             → modules/field/mod_poisson.f90 →
submodules/field/submod_poisson_sor.f90 (or wrapper to select solver)
```

Mapping — ran2.f90

`ran2` → `modules/core/mod_rng.f90` → `submodules/core/submod_rng.f90`

Mapping — read_input.f90

read_input → modules/core/mod_config.f90 → submodules/core/submod_config.f90

Mapping — read_reactions.f90

```
read_reactions                → modules/particles/mod_reactions.f90 → submodules/particles/submod_reactions.f90
ordering                      → modules/particles/mod_reactions.f90 → submodules/particles/submod_reactions.f90
```

Mapping — restart.f90

```
restart                              → modules/io/mod_restart.f90 → submodules/io/submod_restart.f90
restart_OMP                           → modules/io/mod_restart.f90 → submodules/io/submod_restart.f90
```

Mapping — sors.f90

```
sor_rb                              → modules/field/mod_poisson.f90 →
submodules/field/submod_poisson_sor.f90
jacobi                              → modules/field/mod_poisson.f90 →
submodules/field/submod_poisson_jacobi.f90
```

Mapping — sorting.f90

```
part_sorting_OMP            → modules/particles/mod_species.f90 →
submodules/particles/submod_species.f90
part_sorting                → modules/particles/mod_species.f90 →
submodules/particles/submod_species.f90
part_sorting_dom            → modules/particles/mod_species.f90 →
submodules/particles/submod_species.f90
```

Mapping — utils.f90

```
part_moments                  → modules/diag/mod_diagnostics.f90 →
submodules/diag/submod_diagnostics.f90
calc_avg                       → modules/diag/mod_diagnostics.f90 →
submodules/diag/submod_diagnostics.f90
MSTIMER                        → modules/core/mod_timer.f90 → submodules/core/submod_timer.f90
stop_calculation               → modules/core/mod_logger.f90 → submodules/core/submod_logger.f90
```

Mapping — write_data.f90

```
write_data                                → modules/io/mod_io.f90 → submodules/io/submod_io.f90
```